This article has been accepted for inclusion in a future issue of this journal. Content is final as presented, with the exception of pagination.

Unified Brain Network Representation Learning via Adaptive Multimodal Fusion for Alzheimer's Disease Analysis

Hongjie Jiang, Yongcheng Zong, Guoheng Huang, Sibo Qiao, Shanshan Wang, Peipeng Liang, Michael Kwok-Po Ng, *Senior Member, IEEE*, and Shuqiang Wang, *Senior Member, IEEE*

***Abstract*—The fusion of structural and functional brain network analysis has been widely applied in the analysis of various brain diseases, especially in identifying the progression stages of Alzheimer's disease. However, most existing multimodal brain network analysis methods separately construct functional and structural networks, making it difficult to incorporate complementary information from several modalities. To tackle this difficulty, we propose AMFusion(Adaptive Multimodal Fusion), a unified brain network construction framework that jointly learns from functional and structural images, thereby efficiently resolving connectivity and node feature learning problems. Our approach begins with a designed quantization encoder to extract structural features from DTI, while dynamic functional connectivity (FC) is constructed from fMRI. Subsequently, a multilevel brain network fusion module is employed to learn and integrate brain connections, considering interactions across different temporal and spatial scales. Finally, a classifier guides further optimization of the brain network, facilitating multi-class disease classification. The performance of the proposed AMFusion was validated using the authentic dataset from ADNI, and experimental results demonstrated the effectiveness of AMFusion. By considering both connection patterns and node features, our method overcomes the limitations of existing approaches, providing a more comprehensive and robust framework for analyzing and diagnosing AD.**

***Index Terms*—Quantization Encoder, Graph Neural Network, Multimodal Fusion, Brain Network Construction.**

Received 25 July 2025; accepted 31 August 2025. This work was supported in part by the National Key Research and Development Program of China under Grant 2023YFC2506902 and Grant 2024YFE0202900, in part by the National Natural Science Foundations of China under Grant 62172403 and Grant 12326614, in part by the GDSTC: Guangdong and Hong Kong Universities 1+1+1 Joint Research Collaboration Scheme under Grant UICR0800008-24, in part by RGC GRF under Grant 12300125, and in part byt the Joint NSFC and RGC under Grant N-HKU769/21. *(Hongjie Jiang and Yongcheng Zong contributed equally to this work.) (Corresponding author: Shuqiang Wang.)*

Hongjie Jiang, Yongcheng Zong, and Shanshan Wang are with the Shenzhen Institutes of Advanced Technology, Chinese Academy of Sciences, Shenzhen 518055, China.

Guoheng Huang is with the School of Computer Science and Technology, Guangdong University of Technology, Guangzhou 510090, China.

Sibo Qiao is with the The School of Software, Tiangong University, Tianjin 300387, China.

Peipeng Liang is with the Beijing Key Laboratory of Learning and Cognition, School of Psychology, Capital Normal University, Beijing 100048, China.

Michael Kwok-Po Ng is with the Department of Mathematics, Baptist University, Hong Kong.

Shuqiang Wang is with the Shenzhen Institutes of Advanced Technology, Chinese Academy of Sciences, Shenzhen 518055, China, and also with the Department of Mathematics, Baptist University, Hong Kong (e-mail: sq.wang@siat.ac.cn).

Recommended for acceptance by Y. Wang.

Digital Object Identifier 10.1109/TETCI.2025.3619586

I. Introduction

MULTIMODAL brain imaging is of significant importance in the analysis of brain diseases. It is evident that a range of imaging modalities have the capacity to capture multidimensional features, including but not limited to brain structure and function. Consequently, this provides complementary information for the analysis of brain diseases [1]. Taking Alzheimer's disease (AD) as an example, functional magnetic resonance imaging (fMRI) detects significant disruptions in global topological features of functional connectivity, while diffusion tensor imaging (DTI) reveals structural degradation of white matter fiber bundles in regions such as the corpus callosum, collectively demonstrating synergistic abnormalities between functional connectivity and white matter structural integrity [2]. These multimodal information is not only applicable to AD, but can also be extended to other neurological diseases such as schizophrenia and Parkinson's disease [3]. Therefore, it is crucial to develop analytical methods that can integrate multimodal imaging data in order to gain a comprehensive understanding of brain diseases.

Using graph theory analysis methods, brain networks can be conceptualized through a framework that includes nodes and edges, symbolizing the relationships and interactions among various regions of the brain [4]. In brain networks, the nodes denote regions of interest (ROIs), while the edges signify the correlations that exist between different brain regions. FMRI BOLD signals capture the temporal relationships between different brain regions, whereas DTI is utilized to map the actual physical pathways connecting gray matter areas. Functional connectivity (FC) enables the observing of the synchronization patterns within brain activity, while structural connectivity (SC) assesses the anatomical linkages of nerve fibers that span across various brain regions [5], [6]. It is a common consensus that FC and SC offer vital insights for diagnosing brain disorders. According to Fornito's research [7], slight perturbations in the small world features of functional brain networks can lead to the integrity disruption of the cognitive state system in diseases. Hence, relying solely on structural networks is insufficient to fully account for brain functional disorders. Recent studies have

2471-285X © 2025 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission. See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: OAKLAND UNIVERSITY. Downloaded on February 19,2026 at 19:41:57 UTC from IEEE Xplore. Restrictions apply.

This article has been accepted for inclusion in a future issue of this journal. Content is final as presented, with the exception of pagination.

shown that multimodal brain networks can achieve better analysis and diagnostic results compared to single-modality brain networks [8], and combining these two modes to construct brain networks is a promising technology.

Current methods for combining FC and SC can be classified into two main categories, differentiated by their reliance on modal dominance [9]. The first method stands on data fusion strategies, such as principal component analysis (PCA), multi-kernel learning, and multi-view embedding. Integrating the features of structural and functional networks enables the exploration of the balance between global and local efficiency within them. The term "balance" is employed to denote the ratio of local efficiency to global efficiency in brain networks [10]. A large number of experiments have proved that more distinctive results can be obtained compared with using only a single mode [11]. The commonly used strategy for the second method is to use one modality to assist another modality [12]. However, this approach fails to completely harness the complementary information between these dual modalities. J. Huang et al. utilized the inter-regional activity correlations represented by functional brain networks as the nodal features for brain regions in structural brain networks [12]. However, the above fusion is insufficient because it fails to fully leverage the complementary information between the two modalities. This is because the node features of brain regions were not constructed in an end-to-end manner. In addition, most existing studies have overlooked node information and global correlations between brain regions [11], [12], [13], [14], [15]. Although certain studies advocate for the inclusion of node information, the metrics typically computed are merely basic topological attributes, including the mean degree and clustering coefficients. Neglecting potentially significant node information that higher-order methodologies could uncover.

This paper proposes a generative multimodal fusion method rooted in Graph Neural Networks (GNNs). In contrast to the two prevalent techniques, the proposed method focuses more on learning functional and structural modal features, and jointly constructs a unified brain network. By introducing the features of structural modal DTI, functional and structural information are organically integrated to construct a universal brain network analysis framework. By delving deeper into the topological structure and connectivity features of brain networks, more comprehensive information support is provided for the early diagnosis, treatment, and mechanism analysis of these brain diseases [16], [17] like Alzheimer's disease, Parkinson's disease, and autism spectrum disorder.

The principal novelties of this approach are listed below:

1) The proposed methodology is a novel adaptive multilevel brain network construction method that fuses DTI and static and dynamic FC from fMRI. Dynamic FC has been demonstrated to introduce temporal information into multimodal brain networks.
2) The adaptive quantization encoding scheme is designed for DTI feature extraction, and its core mechanism, Mixup strategy, can effectively augment training data. Adaptive weighting is employed to reduce overfitting in small sample training, enabling personalized attention to diverse features and ultimately enhancing the generalization capability of brain network construction.
3) The structural-functional fusion learning module is proposed for multimodal fusion of brain networks, aggregating the cross-temporal correlations among hierarchical graphs. The sorting pooling in the structure-function fusion module learns prior knowledge of structural positions in high-dimensional space, and adaptive message propagation adjusts the importance of different brain regions, thereby improving interpretability of the generated brain network.

II. Related Work

The network design of the AMFusion framework is comprised of three sub-modules: (a) functional connectivity construction and diffusion tensor imaging encoding, (b) structure-function fusion learning, and (c) classifier and brain network analysis. Module (a) employs the sliding window method to construct both static and dynamic FC, utilizing an AQE to achieve preliminary modal fusion during the extraction of structural features from DTI data as FC node features. Module (b) employs GNN and Transformer for further multimodal fusion to construct multimodal brain networks. Module (c) applies the constructed multimodal brain networks to disease analysis and diagnosis. The details can be seen in Fig. 1.

A. Multimodal Fusion Method

For the task of classifying diseases, two frequently utilized approaches are used to learn and fuse potentially complementary information from multimodal data, termed as feature concatenation and feature fusion. In feature concatenation, features from a variety of modalities are combined to form a single feature vector, which serves the downstream classification objectives. Numerous scholars endeavor to integrate features from structural MRI with those from other imaging modalities (like dMRI, fMRI, and PET) to enhance classification results. M. Li et al. [18] combined the area volume metric extracted by structural MRI, the diffusion metric generated by dMRI, and the correlation metric calculated by fMRI as input features for AD classification. X. Tang et al. [19] integrated volume, shape, and dispersion metrics of the hippocampus and amygdala into a high-dimensional feature vector, subsequently applying principal component analysis and a T-test for feature dimensionality reduction. After processing, the features were submitted to an SVM classifier to delineate the various stages of AD. The advantage of feature concatenation is that it is direct and simple to implement, but this strategy faces a considerable restriction: it treats multimodal features with equal consideration, disregarding the distinct characteristics and possible significance of features derived from various modalities. Feature fusion merges features from various modalities by employing intricate techniques to cater to the distinct information encapsulated within each. Yang et al. [20] introduced an integrated brain network construction pipeline that concurrently assimilates information from functional and structural neuroimaging. A correlation model was formulated by them to connect all brain regions with functional imaging

Authorized licensed use limited to: OAKLAND UNIVERSITY. Downloaded on February 19,2026 at 19:41:57 UTC from IEEE Xplore. Restrictions apply.

This article has been accepted for inclusion in a future issue of this journal. Content is final as presented, with the exception of pagination.

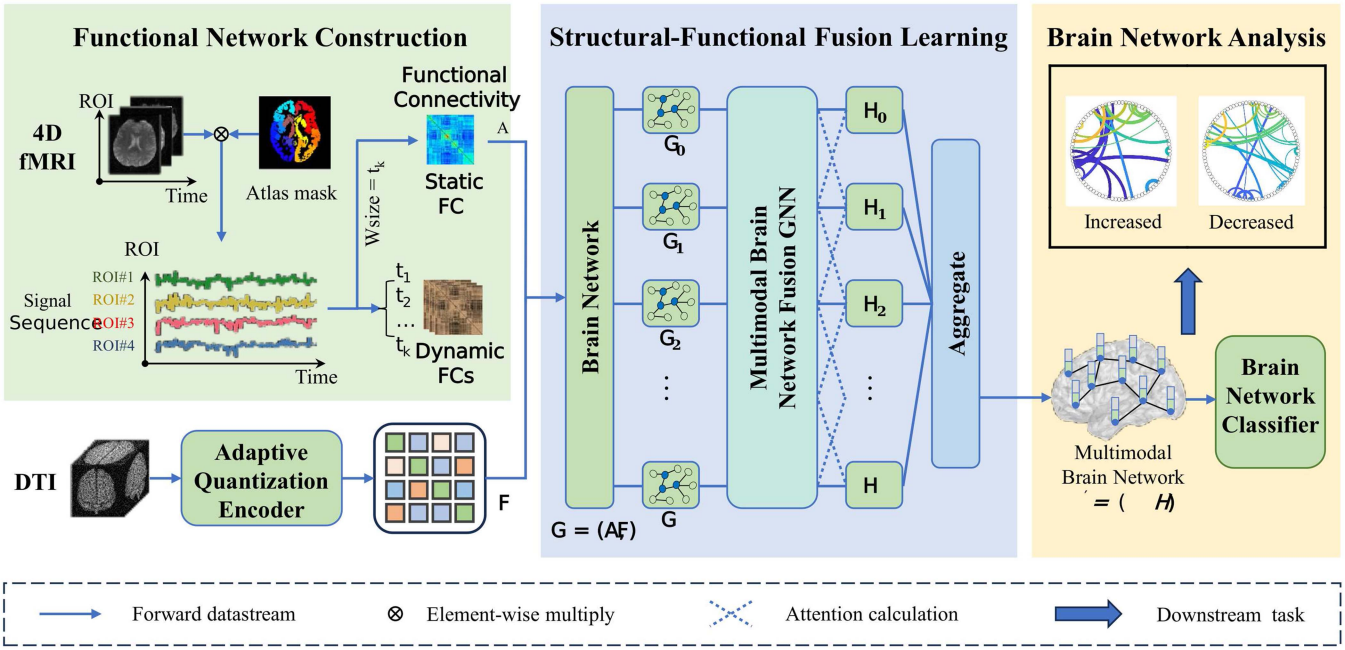


Fig. 1. The AMFusion framework's network design comprises three sub-modules: (a) Functional connectivity construction and DTI encoding, (b) structural-functional fusion learning, and (c) classifier and brain network analysis.

through the method of low-rank representation, and also to integrate local manifolds from structural data into the model at the same time. Q. Zhu et al. fused DTI and fMRI via optimal transport theory combined with a multi-channel spatio-temporal graph convolutional network to extract spatio-temporal topological features of dynamic brain networks [11]. J. Huang et al. construct a brain network with functional connectivity from fMRI as nodes and structural connectivity from DTI as edges, fusing functional and structural connections via attention diffusion and bilinear pooling [12]. Z. Qiu et al. integrate sMRI and PET data through a Transformer-based multi-fusion joint learning module, capturing global inter-modal relationships, local associations, and latent information via global-aware learning, local-aware learning, and outer latent-space learning, respectively, to achieve multimodal fusion [21]. B. Lei et al. integrated genetic, imaging, protein, and clinical data, alleviated heterogeneity through feature induction learning, and employed a dual multi-level graph neural network to extract local information, fuse global information, and obtain multi-level features, thereby optimizing the fusion process [22]. Compared with previous studies, this research introduces end-to-end constructed node information to fully consider inter-modal associations, enabling more sufficient fusion based on connection information. The details can be seen in Fig. 1.

B. Graph Neural Networks in Neuroimaging

Graph neural networks (GNNs) are extensively utilized across various domains within neuroimaging [23], particularly in predicting brain diseases, and have achieved great success [24]. According to the definition of graphs, existing GNN based neuroimaging research can be categorized into two types: population level graphs and individual level graphs. Within a group level graph, every node represents the data for a distinct sample, and the edges represent the similarity between two connected individuals. The individual-level method constructs individual maps per topic. A single node within the graph signifies an associated region of interest (ROI), such as a brain region identified in a brain map, and edges represent the relationship between two ROIs (such as the connectivity between structural brain connections or functional connections).

Y. Zhang et al. [25] formulated multiple graphs per individual subject in a multimodal brain network, trained different GNNs, and connected the features generated by each GNN to classify Parkinson's disease. S. I. Ktena et al. [26] constructed a singular graph based on functional connectivity for each topic and crafted a Siamese Graph Convolutional Network (s-GCN) for acquiring metrics of graph similarity. Within this study, both GNNs have employed a shared weight. In the course of training, two GNNs are fed with two images, originating from the same clinical group and different clinical groups, and the FC layer subsequently providing an estimation of similarity. Generally speaking, most GNN methods optimize the topology of the graph during the full training session. Q. Zhu et al. employed optimal transport theory to capture topological evolution and fused structural and functional brain networks using a multi-channel spatio-temporal graph convolutional network to extract spatio-temporal features [11]. J. Huang et al. constructed brain networks with fMRI functional connections as nodes and DTI structural connections as edges, fusing functional and structural connections via attention diffusion and bilinear pooling [12]. The limitation of graph neural networks in processing brain networks lies in their finite-step diffusion mechanism, which fails to adequately account for long-range interregional correlations.

Authorized licensed use limited to: OAKLAND UNIVERSITY. Downloaded on February 19,2026 at 19:41:57 UTC from IEEE Xplore. Restrictions apply.

This article has been accepted for inclusion in a future issue of this journal. Content is final as presented, with the exception of pagination.

C. Graph Neural Networks in Brain Network

GNN has achieved remarkable research results in various aspects of brain network computing. (1) In terms of brain network classification and prediction, Paris S et al. [27] used a population graph with individual subjects as nodes and the relationships between subjects as edges, combined with the Graph Convolutional Network (GCN) method, to achieve accurate prediction of ASD and AD diseases. J. Ma et al. [28] creatively implemented an attention guided deep GNN for analyzing AD data derived from sMRI, thereby uncovering the most relevant brain regions and basic time points within AD progression. (2) In terms of brain network representation learning, Z. Qiu et al. [21] implemented a Multi-Fusion Joint Learning (MJL) module that effectively integrates multimodal features and enhances feature representation from global, local, and latent learning perspectives. In contrast to methods that concentrate on static brain networks, STAGIN [29] creates dynamic brain networks derived from fMRI data, employing GNN with spatiotemporal attention mechanisms. (3) In terms of multimodal fusion learning, J. Gan et al. [30] crafted a multi graph fusion to investigate shared and supplementary information within functional brain networks. L. Zhang et al. [31] implemented a deep GNN model GBDM that integrates topological features of fMRI and DTI. This model achieves the construction of multimodal deep brain connections and supports the prediction of mild cognitive impairment in early AD and brain network analysis. H. Cui et al. [32] first established a benchmark for multimodal brain network analysis using message passing GCN, providing direction for establishing a unified, scalable, and replicable GNN brain network analysis framework.

III. Methodology

Constructing brain networks is a crucial challenge within the realms of neuroscience and computational neuroscience, aiming to deduce the network topology and functional connectivity patterns among brain regions from neuroimaging datasets. Specifically, brain network construction tools generally take raw neuroimaging data as input, and generate a brain network for each input sample through a series of processes such as data preprocessing, template registration and segmentation, and feature extraction. According to the commonly used AAL brain region [33] partitioning method in neuroscience, the output result is a two-dimensional adjacency matrix $A \in \mathbb{R}^{90\times 90}$ representing the brain network connections.

A. Functional Brain Network Construction

To fully harness the sequential insights contained within fMRI datasets, this paper constructs two functional brain networks (FC), namely static brain networks and dynamic brain networks. Firstly, the 4D fMRI is subjected to standard brain network templates to extract regional signals, and preprocessed to obtain a time series of 90 ROIs. Subsequently, determine the Pearson correlation coefficient among ROIs to establish the static FC. To investigate the dynamics of resting state FC, this paper adopts the frequently applied sliding window method [34]. We choose a fixed-length time window and use the data points within that window to calculate dynamic FC. Window size and stride are two hyperparameters of sliding windows. Set the window size w to vary between 32, 64, and 128, and select the starting point for randomly generating the window. For each generated window, the extracted time series will be used to compute the correlation matrix as the adjacency matrix of the functional network. This greatly increases the diversity of model training data, resulting in a more robust model at the end of training.

Algorithm 1: Brain Network Analysis Process.

Require: X: the processed functional MRI data
Require: DTI: Diffusion Tensor Imaging data
Ensure: A': multimodal brain network matrix
procedure FC Network Construction
  $SFC \leftarrow$ ComputeStaticFC$(fMRI, AAL)$
  $DFC \leftarrow$ ComputeDynamicFC$(fMRI, AAL)$
  return SFC, DFC
end procedure
procedure Adaptive Quantization
  $F \leftarrow$ Quantize(DTI)
  return F
end procedure
procedure Brain Network Representation Learning
  $G \leftarrow$ ConstructGraph(SFC, DFC, F)
  for $k \leftarrow 1$ to K **do**
    $S_k \leftarrow$ GraphConvolution(G_k)
    $S_k \leftarrow$ SoftPooling(S_k)
    $H_k \leftarrow$ MLP(S_k)
  end for
  $H_i \leftarrow$ Attention(Q, K, V)
  $H_{fusion} \leftarrow$ MultiScale(Q, K, V)
  $A' \leftarrow$ Aggregate$(\{H_1, H_2, \ldots, H_k\})$
  return A'
end procedure
procedure ClassifyBrainNetwork
  $y \leftarrow$ Classifier(A')
  return Classification Result y
end procedure
Calculate:
$\mathcal{L}_{cls} = -\sum_{i=0}^{C-1} y_i log\,(p_i) + (1-y_i)\,log\,(1-p_i)$
Update: Optimize the modules
$A' \leftarrow$ BrainNetwork(SFC, DFC, F)
$y \leftarrow$ ClassifyBrainNetwork(A')

Assuming that the time series of a preprocessed individual subject is represented as a matrix X that measures N $\times$ T, where $N = 90$ represents the BOLD signal composition with a longer T time in 90 ROIs. The static functional connection is obtained from the following equation:

$$SFC_{ij} = \frac{\mathrm{Cov}\,(X_i, X_j)}{\sigma_{X_i}\sigma_{X_j}} = \frac{E\left((X_i - \mu_{X_i})\left(X_j - \mu_{X_j}\right)\right)}{\sigma_{X_i}\sigma_{X_j}} \tag{1}$$

For dynamic functional connections, by setting different window sizes w, we obtain a matrix X_w with a size of $N \times w$. Using the same calculation method, we obtain a set of dynamic

Authorized licensed use limited to: OAKLAND UNIVERSITY. Downloaded on February 19,2026 at 19:41:57 UTC from IEEE Xplore. Restrictions apply.

This article has been accepted for inclusion in a future issue of this journal. Content is final as presented, with the exception of pagination.

functional connectivity matrices $\{DFC_1, DFC_2, \ldots, DFC_k\}$. DTI is used to extract node features, and sparse encoding feature $F_{N\times d}$ is obtained through adaptive quantization encoder. Denote $A_0 = SFC_{90\times 90}$, $A_i = DFC_i (1 \le i \le k)$. By using functional connections as edges of the graph and encoding features as nodes of the graph, i.e. $G_i = (A_i, F)$, a multi-layer brain network $G = G_0, G_1, G_2, \ldots, G_k$ is formed.

B. Adaptive Quantization Encoder

The Adaptive Quantization Encoder (AQE) plays a pivotal role in processing and analyzing Diffusion Tensor Imaging data. AQE extracts structural features from DTI as node features of the functional brain network, and collaborates with the downstream structure-function fusion module to achieve full integration of structural and functional information. This enables the constructed multimodal brain network to fully characterize the brain features of AD. Additionally, AQE transforms the continuous structural features of DTI data into discrete layers through quantization operations. This process can highlight the discrete changes in brain region structural connections caused by AD, thereby enhancing the AD brain features. Our AQE leverages a customized lightweight neural network, ShuffleNetV2 [35], as its backbone, selected for its computational efficiency and its ability to adaptively adjust the weight of quantized features. Fig. 1 illustrates a conceptual representation of the enhancement of the convolutional method through the backbone network. Through this adaptive quantization, we have achieved personalized attention to different features and improved the learning ability of neural networks on high-dimensional DTI data.

The core operational principle of the AQE revolves around the adaptive adjustment of the quantization weights, which are tailored based on the complexity and variability of the feature vectors extracted from the DTI scans. This adaptability ensures that more complex regions receive a detailed quantitative representation, enhancing the fidelity and utility of the encoded features for brain network reconstruction.

The AQE incorporates a novel mechanism to adaptively weight quantization-level features by computing their complexity as weights, which are derived using entropy calculation. Dynamic weight adjustment is achieved through Equation (3) in combination with feature-specific entropy values. This enables the Structure-Function Fusion Learning module to prioritize features with higher information content. This is particularly important in capturing the intricate fiber structures in DTI, which are critical for accurate brain connectivity analysis. The quantization process can be mathematically represented as:

$$Q(f_i) = \text{round}\left(\frac{f_i - \min(F)}{\max(F) - \min(F)} \cdot (L-1)\right) \tag{2}$$

where $Q(f_i)$ is the quantized feature, f_i is the i-th feature in the feature set F, and L is the number of quantization levels. The adaptive weights are then applied:

$$w_i = \frac{\exp(-\text{Entropy}(f_i))}{\sum_{j=1}^{N} \exp(-\text{Entropy}(f_j))} \tag{3}$$

where w_i is the adaptive weight for the i-th feature, calculated inversely proportional to its entropy to prioritize features with higher information content.

This research assumes that the reason for the limited diversity and robustness of artificial intelligence-based medical data diagnosis is that too few samples lead to overfitting of model learning. To enhance the diversity and robustness of the training dataset, a Mixup strategy is incorporated during the training phase of AQE. Mixup performs linear interpolation between pairs of randomly selected samples, creating synthetic samples that help the model generalize better across different brain structures and pathological conditions. The Mixup operation is formalized as:

$$\begin{aligned} \tilde{x} &= M \odot x_a + (1-M) \odot x_b \\ \tilde{y} &= \lambda y_a + (1-\lambda) y_b \end{aligned} \tag{4}$$

where M represents a rectangular mask applied in the feature space, $\odot$ represents element-wise multiplication, and λ a mixing coefficient drawn from a β distribution, Beta(α, α), ensuring that the mask's coverage ratio aligns with $\lambda = \frac{\sum M}{HW}$.

C. Structural-Functional Fusion Learning

There are various fusion methods for brain networks [11], [12], [36], [37]. Earlier studies within this domain have investigated diverse techniques for constructing brain networks and subsequent downstream tasks. Using different measurement methods, for instance, average activation, node index as coordinates, spatial one hot encoding, and the most common correlation, can all be used as node features in the graph. In addition, various measurement methods comprising partial correlation, Pearson correlation, and geometric distance have been widely used to define edges in brain images. At this stage, the feature matrix is depicted through $F \in N \times d$, where N represents the overall count of ROIs and d represents the feature vector's dimensions. By utilizing the graph representation capability of GNN and Transformer to model the long-range relationships of time series, the fusion of structural and functional image data is achieved.

$$\begin{aligned} h'^{(l)} &= \text{Aggregate}\left(\text{Norm}(h^{(l-1)}\right) + h^{(l-1)} \\ h^{(l)} &= \text{FFN}\left(\text{Norm}(h'^{(l)}\right) + h'^{(l)} \end{aligned} \tag{5}$$

The multilevel brain network fusion module is divided into two parts: graph optimization and graph aggregation. With three layers of GNN, we use a graph classification architecture as the graph optimization learning module, subsequently a sorting pooling aggregator. Sorting pooling sorts node features according to the final channel, retaining solely the top k depictions. Thereafter, sorting pooling is refined through the integration of two one-dimensional convolutional layers, ultimately employing a two-layer multi-layer perceptron (MLP) to map features into a constant dimension. The graph aggregation module Transformer integrates hidden layer features from multiple time series and learns cross temporal correlations between different levels of graphs. With multi-head attention, the model gains the ability to zero in on information from different time scales at different

Authorized licensed use limited to: OAKLAND UNIVERSITY. Downloaded on February 19,2026 at 19:41:57 UTC from IEEE Xplore. Restrictions apply.

This article has been accepted for inclusion in a future issue of this journal. Content is final as presented, with the exception of pagination.

positions, that is, through MultiHead, the model can capture richer feature information.

$$\begin{aligned} H_i &= \text{Attention}\left(Q_i, K_i, V_i\right) \\ &= \text{softmax}\left(\frac{Q_i\left(K_i\right)^T}{\sqrt{d}}\right) V_i^F \end{aligned} \tag{6}$$

$$\begin{aligned} H_{fusion} &= \text{MultiScale}(Q, K, V) \\ &= \text{concat}\left(H_0, H_1, \ldots, H_h\right) \end{aligned} \tag{7}$$

where $Q = S_i W_i^Q, K = S_i W_i^K, V = S_i W_i^V$.

Finally, use Readout concatenation as the embedding for aggregating individual graphs, calculate the feature mean between each vertex, multiply them by their respective weights, and add up the results obtained for each node.

$$h_i^{(l+1)} = \delta\left(\sum_{j \in S} A^{(l)} h_j^{(l)}\right) \tag{8}$$

D. How Modules are Connected

We extract DTI features with an AQE module, attach them as node features to an fMRI-derived functional network, fuse them via a GNN and then a Transformer, and obtain a multimodal brain network by computing inner products of the final node embeddings for downstream analysis.

E. Classifier and Brain Network Analysis

We designed a multi-classification classifier to divide the multimodal brain networks. Within this module, features are aggregated by an MLP before being finally connected to the Softmax function, which outputs the estimated probabilities for each disease classification. Specifically, the classifier comprises three layers of backpropagation neural networks, two layers of ReLU activation functions, and a Softmax output layer. The Softmax function takes C neurons as input and maps the probability of the results to four disease categories. Multi-class loss adopts a multi-class cross entropy function. In AD disease prediction, the number of categories is 4.

$$\mathcal{L}_{cls} = -\sum_{i=0}^{C-1} y_i \log\left(p_i\right) + \left(1 - y_i\right) \log\left(1 - p_i\right) \tag{9}$$

Meanwhile, we will compare the multimodal brain network with the initial static network to study changes in brain connectivity and visualize important brain region information. To ensure symmetry, the multimodal brain network is determined by the following formula. In the following experimental section, we will delve into the results of constructing brain networks.

$$A' = \sigma(H^T H) \tag{10}$$

H refers to the features obtained by the Structural-Functional Fusion Learning module through the fusion of multimodal information. H^T refers to the transposition of features H. $\sigma(\cdot)$ refers to the Sigmoid function, which is used to control the value range of edge weights. A' is the adjacency matrix of the obtained multimodal brain network, and the formula (10) ensures its symmetry.

TABLE I
SUBJECTS' DETAILS OF THIS STUDY

Group	NC(85)	EMCI(120)	LMCI(60)	AD(60)
Gender	44M/41F	63M/57F	26M/34F	31M/29F
Age	74.3±5.5	74.9±5.8	75.8±6.1	75.6±5.4

NC: Normal Control, EMCI: Early Mild Cognitive Impairment, LMCI: Late Mild Cognitive Impairment, AD: Alzheimer's Disease, M: Male, F: Female.

IV. EXPERIMENTS

A. Dataset and Preprocessing

This study chose fMRI and DTI data for constructing multimodal brain networks. This is because these two types of data contain the brain's functional information and the structural information of white matter fibers, respectively. The DTI and fMRI imaging data were entirely obtained from the Alzheimer's Disease Neuroimaging Initiative (ADNI) [38] public data repository for the verification of our suggested framework. This research incorporated data from 325 subjects. A comprehensive summary of subject details is presented in Table 1.

This study used the PANDA [39] toolbox for raw DTI data preprocessing. Firstly, convert the original DICOM-formatted DTI data to NIFTI format data. Then perform skull dissection, fiber bundle resampling, and head motion correction. Subsequently, we compute the fractional anisotropy (FA) coefficient through the tensor model optimization via the least squares method and output the DTI data. After resampling, the voxel dimensions for all subjects' DTI were standardized to a $91 \times 109 \times 91$ matrix, with each voxel measuring $2\,\text{mm} \times 2\,\text{mm} \times 2\,\text{mm}$. Next, we will construct a functional brain network for fMRI. Choose AAL3.0 as the brain template, comprising 90 ROIs, with each delineated as a brain network node. Preprocess using GRETNA to obtain a time series of 90 ROIs. For DTI Noise Elimination via PANDA, head motion correction was performed to rectify artifacts caused by subjects' head movements during scanning, ensuring more accurate spatial positioning of the data. Fiber bundle resampling standardized data acquisition to reduce noise introduced by irregular sampling. Voxel-wise dimension normalization was applied to unify all subjects' DTI data into a fixed matrix size, regulating data in terms of morphology and spatial position to minimize noise interference.

The standard procedure from the GRETNA toolbox [40] was utilized for the initial processing of fMRI data in the experiment. Initially, the functional time series signal undergoes filtration, with principal stages comprising magnetization balance calibration, magnetic head motion artifact correction, spatial normalization, and bandpass filtering within the range of 0.01 Hz and 0.08 Hz. Then, 90 non-overlapping ROIs of brain regions were parcellated using the Automatic Anatomical Labeling (AAL) template. Finally, we normalized the time series signals to the same length, obtaining a matrix for each participant with the size of 90×187, indicative of the functional time series X. For fMRI Noise Elimination via GRETNA, band-pass filtering

Authorized licensed use limited to: OAKLAND UNIVERSITY. Downloaded on February 19,2026 at 19:41:57 UTC from IEEE Xplore. Restrictions apply.

This article has been accepted for inclusion in a future issue of this journal. Content is final as presented, with the exception of pagination.

TABLE II
RESULTS OF ABLATION STUDY ON EACH BLOCK PROPOSED

AQE	MBNF	AUC	ACC	SEN	SPE
×	×	72.14	71.05	79.99	67.86
×	✓	78.93	81.58	60.00	89.29
✓	×	76.30	82.05	55.55	90.00
✓	✓	83.50	84.21	63.64	92.59

was first used to remove high-frequency and low-frequency interferences, retaining effective frequency signals associated with brain activities. Spatial normalization was employed to unify the data spatial standard, while magnetic head motion artifact correction addressed noise from head movements during scanning. Detrending and other operations were conducted to eliminate slow linear or nonlinear drifts in the data.

B. Experiment Settings

This model undergoes training and testing on the PyTorch framework, utilizing NVIDIA RTX 4000 GPU hardware along with 20 GB of graphics memory. Throughout the training phase, the optimization algorithm is configured as Adam, initiated with a starting learning rate of 0.0001, experiencing exponential decay relative to the training iteration count. The number of epochs is 400, and the batch size is configured for a total of 32. The experiment adopts a five-fold cross-validation to assess the model's efficacy. Five subsets of the same size were divided from all participants. The role of one subset is designated for testing, and the aggregate of the other four is for training purposes. Execute this procedure five times, ensuring unbiasedness via random partitioning. The indicators for evaluating classification effectiveness in this experiment are accuracy (ACC), sensitivity (SEN), specificity (SPE), and ROC curve area (AUC).

C. Ablation Study and Hyperparameter Analysis

The ablation experiment is used to demonstrate the role of the feature extraction module AQE and the multi-level brain network fusion module MBNF. Replace the DTI feature extraction module with a regular MLP, and replace the brain network fusion module with a fixed scale brain network input and a regular message passing graph neural network lacking attention layers to complete the ablation experiment.

From Table II, it is observable that the two implemented modules can effectively improve model performance in all indicators. Especially the brain network fusion module MBNF, compared to AQE, has a more significant improvement in model performance.

Within the suggested AMFusion framework, the feature dimension hyperparameter d of brain images in the AQE encoder controls the classification performance and training cost, and it is necessary to balance the selection of the best input to reduce the dimensionality in the subsequent fusion model. We set d to 32, 64, 128, 256, 512, and then apply five-fold cross-validation to determine the optimal hyperparameter, and display the classification results in a violin plot. Fig. 2 illustrates the ACC of the model's classification performance across various hyperparameter settings. It is observable that at d=128, the model realizes optimal classification performance. It should be highlighted that the model's performance declines if the value of d is excessively high or low. When d is too small, the embedded features contain too little information. When d is too large, our model may experience overfitting, resulting in poor generalization on the test set.

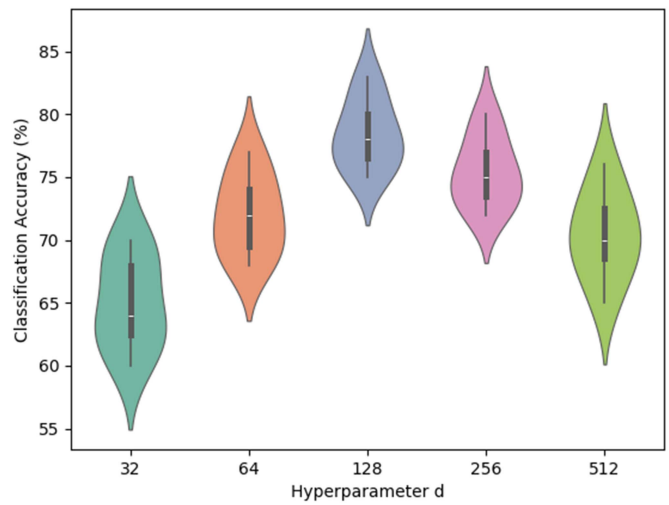


Fig. 2. The effect of the hyperparameter d on classification accuracy.

TABLE III
PREDICTIVE PERFORMANCE INDICATORS UNDER DIFFERENT CATEGORIES ON ADNI DATASETS

categories	AUC	ACC	SEN	SPE
NC	83.23	75.07	89.61	70.21
EMCI	82.64	79.93	70.37	86.00
LMCI	86.15	82.23	84.61	81.75
AD	91.13	92.55	85.51	94.02

V. DISCUSSION

This study conducts diagnostic experiments for Alzheimer's disease, analyses of brain connectivity changes, and graph-theoretical index analyses based on the constructed multi-modal brain network. It is found that the multi-modal brain network constructed by the framework in this study can not only effectively distinguish different stages of Alzheimer's disease, but also identify pathogenic circuits, pathological brain regions, and changes in graph-theoretical indices. This demonstrates the rationality and effectiveness of the multi-modal brain network constructed by the framework.

A. Classification Performance

The model's outcomes on the test dataset were validated through 5-fold cross validation, with an average classification ACC of 0.82. The model achieved good performance on the test set. On the four binary classification tasks of NC, EMCI, LMCI, and AD (one vs. rest), the accuracy of the model reached 75.07%, 79.93%, 82.23%, and 92.55%, respectively. The AUC values were also above 82%, indicating that the fused brain network can effectively capture the feature differences of different disease

Authorized licensed use limited to: OAKLAND UNIVERSITY. Downloaded on February 19,2026 at 19:41:57 UTC from IEEE Xplore. Restrictions apply.

This article has been accepted for inclusion in a future issue of this journal. Content is final as presented, with the exception of pagination.

TABLE IV
A QUANTITATIVE EVALUATION OF THE PREDICTION PERFORMANCE(%) OF A RANGE OF METHODS ON ADNI DATASETS

Method	AUC	ACC	SEN	SPE
GAT	76.63	81.95	69.83	**85.86**
GIN	82.82	76.32	**88.75**	72.93
OT-MCSTGCN	84.38	81.25	75.00	**87.50**
Ours	**85.79**	**82.45**	82.53	83.00

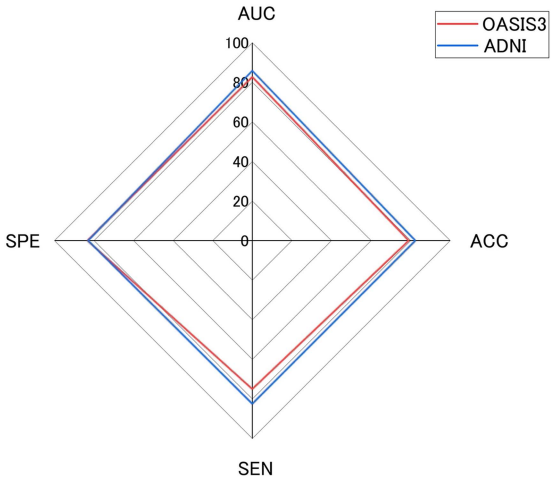
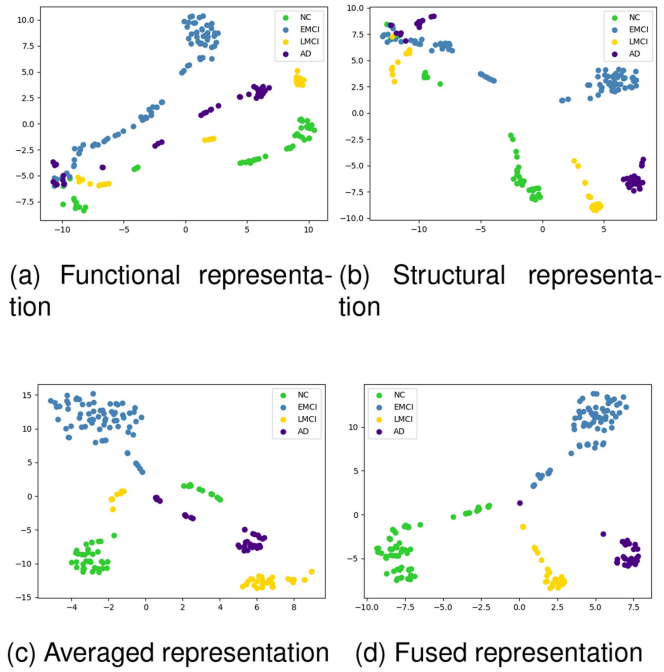


Fig. 3. The validation of generalization ability on ADNI and OASIS3 datasets.



(a) Functional representation (b) Structural representation

(c) Averaged representation (d) Fused representation

Fig. 4. The dimensionality reduction visualization of brain network representations obtained through four different brain network construction methods. Panels (a) and (b) represent the classification features captured by GRETNA and PANDA software, respectively, while panels (c) and (d) represent the classification features captured by the multi-modal brain network constructed using simple weighting and the proposed method. Green dots represent NC, blue dots represent EMCI, yellow dots represent LMCI, and purple dots represent AD.

stages. Among them, the model performed the best on AD with an AUC of 91.13%, and its sensitivity and specificity reached 85.51% and 94.02%, indicating that the optimized brain network has high discriminative ability for AD stages.

For a comprehensive comparison, we introduced two cutting-edge graph neural network models, namely OT-MCSTGCN [11], Graph Attention Transformer (GAT) [41] and Graph Isomorphism Network (GIN) [42], and obtained three baseline methods, respectively. As illustrated in TABLE IV, the proposed method exhibits superior performance in the quantitative assessment of prediction capabilities on the ADNI dataset. A comparison of our model with GAT, GIN, and OT-MCSTGCN reveals that it exhibits superior performance in terms of AUC and ACC. The AUC value of 85.79% surpasses those of 76.63% of GAT, 82.82% of GIN, and 84.38% of OT-MCSTGCN, thereby underscoring the model's superior capacity to differentiate between normal and AD patient samples. The ACC of 82.45% obtained in our model surpassed the performance of other methods, thereby substantiating the efficacy of our model in accurately categorizing AD patients and normal samples. Meanwhile, the SEN of Ours yielded a result of 82.53%, while the SPE registered at 83.00% . These findings suggest that the method demonstrates efficacy in identifying AD-positive cases and excluding normal-negative samples. The rationale underlying this choice is twofold. Firstly, compared with comparative methods, our model constructs node features end-to-end and adopts a multi-level fusion architecture. Secondly, It not only captures inter-modal correlations, but also captures temporal correlations via dynamic FC.

This research utilized 50 cognitively normal adults and 50 mild cognitive impairment patients from OASIS3 [43] to assess the generalizability of our model. Fig. 3 clearly demonstrates the performance of our model across different datasets. A comparison of the model's performance on the OASIS3 and ADNI datasets reveals that the performance curves are analogous, exhibiting a mere 3.15% discrepancy in AUC. The observed cross-dataset performance consistency indicates that the model exhibits effective adaptability to varying data distributions and demonstrates superior generalizability.

This research further analyzed the differences between different types of brain networks through feature visualization, and used the t-SNE tool [44] to graphically represent the mapping of classification features onto a 2D space. As shown in Fig. 4, compared with software tools and simple weighted brain networks, the classification features extracted by AMFusion provide better discriminative clustering. At the same time, it can be seen that using a single modal brain network will lead to points that are difficult to distinguish, while the fusion network can distinguish different categories well.

B. Brain Network Construction Results

Public connections are defined as brain connections of equal strength in both functional and structural brain networks, while private connections are significant connections that only exist in one of these networks. The fusion brain network can comprehensively learn the public and private connections between the two.

In Fig. 5, it can be seen that the fused brain connections exhibit a pattern of two modal connections, which is not simply the superposition of two networks, but a comprehensive feature expression. The fused connection modality not only reflects

Authorized licensed use limited to: OAKLAND UNIVERSITY. Downloaded on February 19,2026 at 19:41:57 UTC from IEEE Xplore. Restrictions apply.

This article has been accepted for inclusion in a future issue of this journal. Content is final as presented, with the exception of pagination.

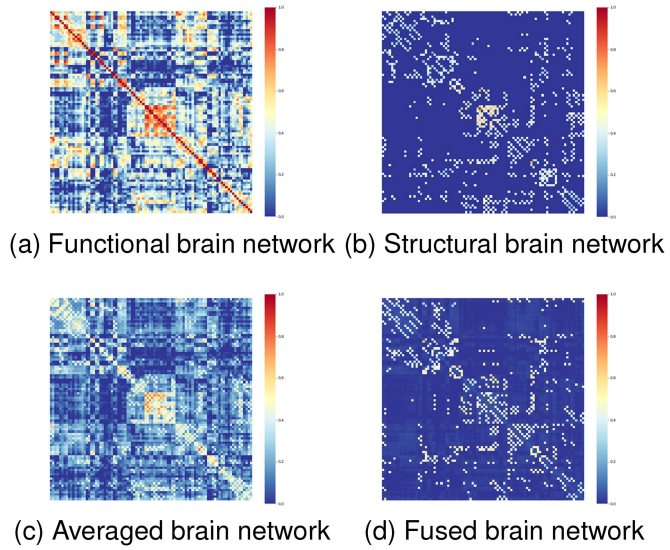


Fig. 5. Comparison of fused brain networks with structural and functional brain networks.

the dynamic activity of functional brain networks, but also considers the connection topology of structural brain networks. This comprehensive feature enables the fused brain connections to more comprehensively describe the correlation that binds brain function with its structure. The fused brain connections may also show some new correlations that cannot be observed in a single mode. These associations may be interactions or regulatory relationships between function and structure, delivering profound insights into the sophisticated interplay of brain function and structure.

C. Changes in Brain Connectivity

In order to quantitatively analyze the characteristics of brain connectivity changes in different stages of cognitive diseases, the differences in connectivity of brain networks in different disease stages were studied based on the structural brain connectivity matrix output by the model. Graphically represent the fluctuations in brain connectivity across various phases. The visualization is shown in Fig. 6, which shows the differences in brain connectivity between the NC vs. EMCI, NC vs. LMCI, NC vs. AD groups. For every case of binary classification task, individually represent the connectivity matrix for each group, subsequently deduct the NC group's connectivity matrix from those of other subject categories to ascertain the shifts in brain network connectivity.

These abnormal brain connections can reflect the damage to important brain connections during the progression of Alzheimer's disease. The brain regions involved in the abnormal increase of EMCI and LMCI connectivity include the cingulate gyrus and occipital lobe, while in the AD stage, the connectivity between the frontal and temporal lobes also increases. Abnormal brain connectivity is concentrated in the left cingulate gyrus and left occipital lobe. For decreased connectivity, all three stages of the disease are concentrated interfacing the frontal lobe with diverse brain regions. According to relevant research, as AD progresses [45], brain networks will gradually lose their small-world attributes, significantly reduce local connections, and shift towards more regular network topologies. In comparison to the healthy control group, MCI and AD patients demonstrated reduced connectivity in the temporal gyrus (TPOmid. L) of the DMN, while connectivity in the intra frontal gyrus (ORBsup. R) and intra frontal gyrus (ORBmid. R) of the DMN appeared in stages (b) and (c) of the figure.

All brain region names and abbreviations related to AAL parcellation used in the following text can be found in [33].

D. Analysis of Important Brain Regions

Aiming to accurately locate brain regions highly correlated with the disease, this study conducted in-depth analysis of brain regions corresponding to significantly enhanced and significantly weakened brain connections during the progression of AD. We defined the consistently present common brain regions corresponding to the enhanced or weakened brain connections from NC to EMCI, LMCI, and AD as important brain regions. We visualized 7 brain regions related to AD, as depicted in Fig. 7. These important brain regions include the left anterior cingulate gyrus and paracingulate gyrus (ACG. L), right middle temporal gyrus (MTG. R), right putamen lenticularis (PUT. R), left orbitofrontal gyrus (ORBsup. L), left superior occipital gyrus (SOG. L), left lingual gyrus (LING. L), and right lingual gyrus (LING. R).

These regions are mostly consistent with previous studies, demonstrating that our model captures important brain regions related to AD. ROIs are mostly found within the frontal, temporal, and occipital lobes [46], [47]. The frontal lobe is primarily associated with advanced psychological activities, comprising physical activity control, language expression, self-awareness, and emotional expression [48]. Visual and linguistic information is memorized by the temporal lobe, with abnormal levels of tau protein in the inferior temporal gyrus in AD patients [49]. Overall, the important ROIs and anomalous connections derived from this model can demonstrate the main features related to AD.

E. Analysis of Brain Network Indicators

In this study, we evaluated the capabilities of the proposed AMFusion framework for constructing brain networks. We utilized four metrics to measure the topological properties of both real and generated macro-networks. These metrics provided insights into the network characteristics and their impact on disease diagnosis. The metrics used in our analysis were as follows [50]:

Characteristic path length (CPL): The characteristic path length metric indicates the average distance between any two vertices in the network. It provides an indication of the network's information transfer ability. A shorter characteristic path length implies more efficient information propagation in the network.

Characteristic path length (CPL): The characteristic path length metric is defined as the average distance between any two vertices in the network.

Authorized licensed use limited to: OAKLAND UNIVERSITY. Downloaded on February 19,2026 at 19:41:57 UTC from IEEE Xplore. Restrictions apply.

This article has been accepted for inclusion in a future issue of this journal. Content is final as presented, with the exception of pagination.

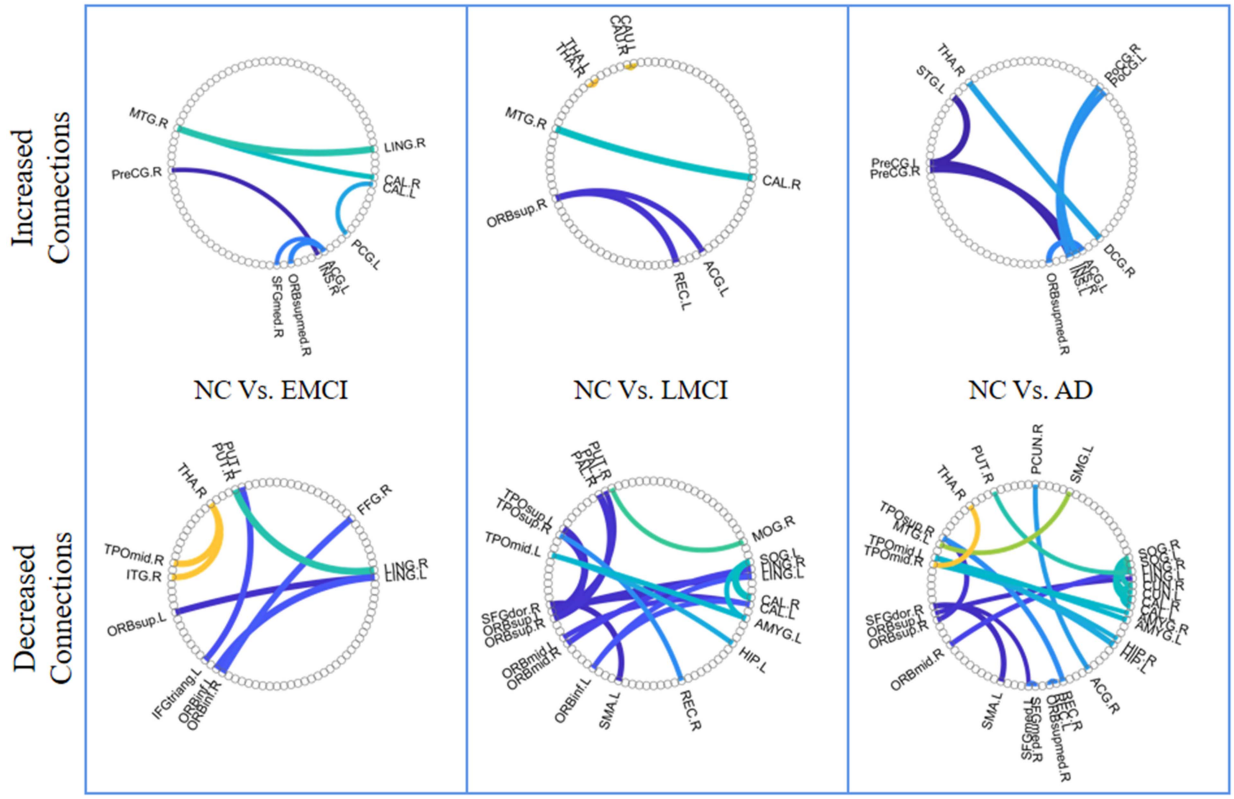


Fig. 6. Shifts in brain network connectivity at multiple stages. The initial row displays the brain connections exhibiting an uptick in connectivity, whereas the following row illustrates those with diminished connectivity. There are significant differences in the brain regions with increased abnormal connections from EMCI to LMCI and AD stages.

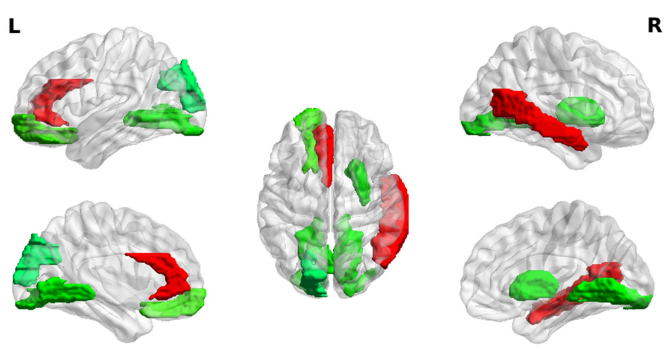


Fig. 7. Visualization of important brain regions related to AD detected by the model. Red represents significantly enhanced brain regions, while green represents significantly weakened brain regions.

Clustering coefficient (CC): The clustering coefficient is a metric that quantifies the degree to which nodes within a network are aggregated.

Global efficiency (GE): The global efficiency metric is a quantitative measure of the network's overall transmission capability of information.

Local efficiency (LE): This index quantifies the efficacy with which information is disseminated within local neighborhoods or clusters of nodes within a network.

For the constructed brain network, different thresholds were used for binarization, and then various brain network indicators were calculated.Finally, the calculated results for all samples

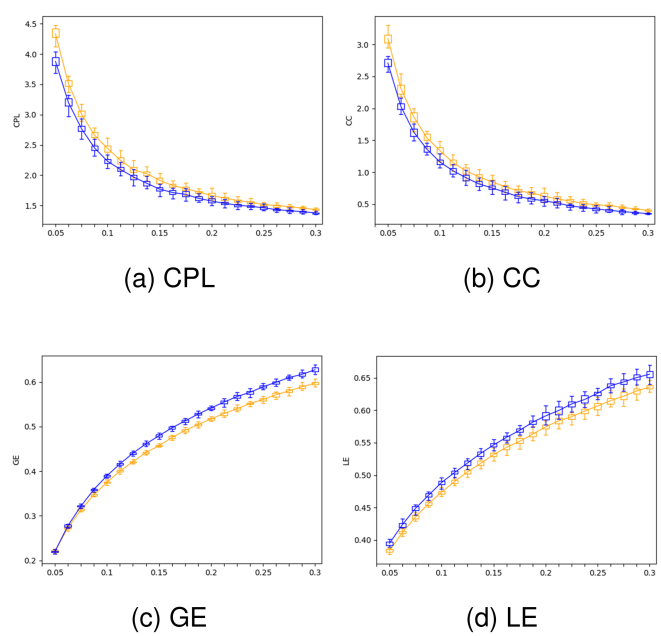


Fig. 8. Brain network analysis in NC(blue) and AD(orange) populations on multiple indicators, from left to right, are CPL, CC, GE, and LE, respectively. The horizontal coordinate of each subgraph represents the threshold selected for binarization of the brain network, the vertical coordinate is the result in the corresponding dimension, and the box plot is used to represent the mean, upper quartile and lower quartile in each group.

Authorized licensed use limited to: OAKLAND UNIVERSITY. Downloaded on February 19,2026 at 19:41:57 UTC from IEEE Xplore. Restrictions apply.

This article has been accepted for inclusion in a future issue of this journal. Content is final as presented, with the exception of pagination.

are presented in Fig. 8. It can be seen that with the increase of threshold, AD and NC show the same trend of change, but the results are significantly different in the four indicators. The brains of AD groups were reduced in both global and local efficiency, which was also reflected in the growth of feature path length. Relevant studies [51] have shown that as cognitive disorders worsen, extensive damage to brain connections leads to a decline in the efficiency of information transmission.

Overall, the experimental outcomes substantiated the efficacy of the suggested AMFusion framework for brain network construction. The metrics employed in the analysis provided valuable insights into the topological properties of the networks and their relevance to disease diagnosis. The findings supported the notion that integrating functional and structural brain network information improves the accuracy and performance of disease classification.

VI. Conclusion

This paper proposes a new brain network construction method AMFusion, which integrates transformer encoder to extract multi-level connectivity features from both structural and functional neuroimaging data. The efficacy of the suggested model was confirmed through the construction of a brain network using the real ADNI dataset, and the experimental findings confirmed the method's potency in subsequent disease classification tasks. This research underscores the significant potential of brain network research, notably its essential function in the analysis of brain networks and the diagnosis of diseases. This work delivers potential insights for the management of multimodal brain imaging datasets and offers an adaptive and powerful approach for the study of brain networks and neuroimaging. First, the suggested framework is yet unable to elucidate the pathogenesis of the identified abnormal brain connection; second, the relatively constrained scope of the dataset employed in this study. In subsequent research, we will improve the brain network module to build a new backbone network and attempt further exploration of brain networks, for instance, investigating the development stages of basic neural circuits and cognitive impairment. Meanwhile, we could verify the efficacy of the suggested framework using more extensive medical image datasets, like the U.K. Biobank.

References

[1] K. Saltoun, B. T. T. Yeo, L. Paul, J. Diedrichsen, and D. Bzdok, "Longitudinal changes in brain asymmetry track lifestyle and disease," *Nature Commun.*, vol. 16, no. 1, 2025, Art. no. 5611.

[2] M. Filippi et al., "Changes in functional and structural brain connectome along the Alzheimer's disease continuum," *Mol. Psychiatry*, vol. 25, no. 1, pp. 230–239, Jan. 2020.

[3] J. Sui, R. Huster, Q. Yu, J. M. Segall, and V. D. Calhoun, "Function-structure associations of the brain: Evidence from multimodal connectivity and covariance studies," *NeuroImage*, vol. 102, no. 1, pp. 11–23, 2014.

[4] B. C. Bernhardt, L. Bonilha, and D. W. Gross, "Network analysis for a network disorder: The emerging role of graph theory in the study of epilepsy," *Epilepsy Behav.*, vol. 50, pp. 162–170, 2015, doi: 10.1016/j.yebeh.2015.06.005.

[5] S. Qi, S. Meesters, K. Nicolay, B. M. ter Haar Romeny, and P. Ossenblok, "The influence of construction methodology on structural brain network measures: A review," *J. Neurosci. Methods*, vol. 253, pp. 170–182, 2015, doi: 10.1016/j.jneumeth.2015.06.016.

[6] Z. Wang, J. Xin, Z. Wang, Y. Yao, Y. Zhao, and W. Qian, "Brain functional network modeling and analysis based on fMRI: A systematic review," *Cogn. Neurodynamics*, vol. 15, no. 3, pp. 389–403, 2021.

[7] A. Fornito, A. Zalesky, and E. T. Bullmore, Eds., *Acknowledgments*. San Diego, CA, USA: Academic Press, 2016, doi: 10.1016/B978-0-12-407908-3.09992-1.

[8] X. Song, A. Frangi, X. Xiao, J. Cao, T. Wang, and B. Lei, "Integrating similarity awareness and adaptive calibration in graph convolution network to predict disease," in *Proc. Med. Image Comput. Comput. Assist. Interv.–MICCAI 2020*, A. L. Martel, P. Abolmaesumi, D. Stoyanov, D. Mateus, M. A. Zuluaga, S. K. Zhou, D. Racoceanu, and L. Joskowicz, Eds., Cham, Switzerland: Springer International Publishing, 2020, pp. 124–133.

[9] J. Huang, L. Zhou, L. Wang, and D. Zhang, "Integrating functional and structural connectivities via diffusion-convolution-bilinear neural network," in *Proc. Int. Conf. Med. Image Comput. Comput.- Assist. Intervention*, 2019, pp. 691–699.

[10] J. D. Rudie et al., "Altered functional and structural brain network organization in autism," *NeuroImage, Clin.*, vol. 2, pp. 79–94, 2013, doi: 10.1016/j.nicl.2012.11.006.

[11] Q. Zhu et al., "Spatio-temporal graph hubness propagation model for dynamic brain network classification," *IEEE Trans. Med. Imag.*, vol. 43, no. 6, pp. 2381–2394, Jun. 2024, doi: 10.1109/TMI.2024.3363014.

[12] J. Huang, L. Zhou, L. Wang, and D. Zhang, "Attention-diffusion-bilinear neural network for brain network analysis," *IEEE Trans. Med. Imag.*, vol. 39, no. 7, pp. 2541–2552, Jul. 2020.

[13] Y. Yang et al., "Brainmass: Advancing brain network analysis for diagnosis with large-scale self-supervised learning," *IEEE Trans. Med. Imag.*, vol. 43, no. 11, pp. 4004–4016, Nov. 2024.

[14] Z. Dong et al., "Brain-jepa: Brain dynamics foundation model with gradient positioning and spatiotemporal masking," in *Proc. 38th Int. Conf. Neural Inf. Process. Syst.*. Red Hook, NY, USA: Curran Associates Inc., 2025, pp. 86048– 86073.

[15] L. Liu et al., "H-net: Heterogeneous neural network for multi-classification of neuropsychiatric disorders," *IEEE J. Biomed. Health Inform.*, vol. 28, no. 9, pp. 5509–5518, Sep. 2024.

[16] T. Huang et al., "Artificial intelligence for medicine: Progress, challenges, and perspectives," *Innov. Med.*, vol. 1, no. 2, 2023, Art. no. 100030.

[17] S. Xu, Y. Liu, H. Lee, and W. Li, "Neural interfaces: Bridging the brain to the world beyond healthcare," *Exploration*, vol. 4, no. 5, 2024, Art. no. 20230146.

[18] M. Li, Y. Qin, F. Gao, W. Zhu, and X. He, "Discriminative analysis of multivariate features from structural MRI and diffusion tensor images," *Magn. Reson. Imag.*, vol. 32, no. 8, pp. 1043–1051, 2014.

[19] X. Tang, Y. Qin, J. Wu, M. Zhang, W. Zhu, and M. I. Miller, "Shape and diffusion tensor imaging based integrative analysis of the hippocampus and the amygdala in Alzheimer's disease," *Magn. Reson. Imag.*, vol. 34, no. 8, pp. 1087–1099, 2016.

[20] J. Yang, Q. Zhu, R. Zhang, J. Huang, and D. Zhang, "Unified brain network with functional and structural data," in *Proc. Med. Image Comput. Comput. Assist. Interv.–MICCAI, 23rd Int. Conf.*, Lima, Peru, Springer, 2020, pp. 114–123.

[21] Z. Qiu et al., "3D multimodal fusion network with disease-induced joint learning for early Alzheimer's disease diagnosis," *IEEE Trans. Med. Imag.*, vol. 43, no. 9, pp. 3161–3175, Sep. 2024, doi: 10.1109/TMI.2024.3386937.

[22] B. Lei et al., "Alzheimer's disease diagnosis from multi-modal data via feature inductive learning and dual multilevel graph neural network," *Med. Image Anal.*, vol. 97, 2024, Art. no. 103213, doi: 10.1016/j.media.2024.103213.

[23] C. You et al., "Artificial intelligence in breast imaging: Current situation and clinical challenges," *Exploration*, vol. 3, no. 5, 2023, Art. no. 20230007.

[24] Y. Liu, Y. Chen, and L. Han, "Bioinformatics: Advancing biomedical discovery and innovation in the era of Big Data and artificial intelligence," *Innov. Med.*, vol. 1, no. 1, 2023, Art. no. 100012.

[25] Y. Zhang, L. Zhan, W. Cai, P. Thompson, and H. Huang, "Integrating heterogeneous brain networks for predicting brain disease conditions," in *Proc. Med. Image Comput. Comput. Assist. Interv.–MICCAI, 22nd Int. Conf.*, Shenzhen, China, 2019, pp. 214–222.

[26] S. I. Ktena et al., "Metric learning with spectral graph convolutions on brain connectivity networks," *NeuroImage*, vol. 169, pp. 431–442, 2018, doi: 10.1016/j.neuroimage.2017.12.052.

[27] S. Parisot et al., "Disease prediction using graph convolutional networks: Application to autism spectrum disorder and Alzheimer's disease," *Med. Image Anal.*, vol. 48, pp. 117–130, 2018.

Authorized licensed use limited to: OAKLAND UNIVERSITY. Downloaded on February 19,2026 at 19:41:57 UTC from IEEE Xplore. Restrictions apply.

This article has been accepted for inclusion in a future issue of this journal. Content is final as presented, with the exception of pagination.

[28] J. Ma, X. Zhu, D. Yang, J. Chen, and G. Wu, "Attention-guided deep graph neural network for longitudinal alzheimer's disease analysis," in *Proc Med. Image Comput. Comput. Assist. Interv.–MICCAI, 23rd Int. Conf.*, Lima, Peru, Springer, 2020, pp. 387–396.

[29] B. -H. Kim, J. C. Ye, and J. -J. Kim, "Learning dynamic graph representation of brain connectome with spatio-temporal attention," in *Proc. 35th Int. Conf. Neural Inf. Process. Syst.*, in NIPS '21, Red Hook, NY, USA: Curran Associates Inc., 2021, pp. 4314–4327.

[30] J. Gan, Z. Peng, X. Zhu, R. Hu, J. Ma, and G. Wu, "Brain functional connectivity analysis based on multi-graph fusion," *Med. Image Anal.*, vol. 71, 2021, Art. no. 102057, doi: 10.1016/j.media.2021.102057.

[31] L. Zhang et al., "Deep fusion of brain structure-function in mild cognitive impairment," *Med. Image Anal.*, vol. 72, 2021, Art. no. 102082, doi: 10.1016/j.media.2021.102082.

[32] H. Cui et al., "Braingb: A benchmark for brain network analysis with graph neural networks," *IEEE Trans. Med. Imag.*, vol. 42, no. 2, pp. 493–506, Feb. 2023.

[33] E. T. Rolls, C. -C. Huang, C. -P. Lin, J. Feng, and M. Joliot, "Automated anatomical labelling atlas 3," *Neuroimage*, vol. 206, 2020, Art. no. 116189, doi: 10.1016/j.neuroimage.2019.116189.

[34] S. Shakil, C. -H. Lee, and S. D. Keilholz, "Evaluation of sliding window correlation performance for characterizing dynamic functional connectivity and brain states," *Neuroimage*, vol. 133, pp. 111–128, 2016, doi: 10.1016/j.neuroimage.2016.02.074.

[35] N. Ma, X. Zhang, H. -T. Zheng, and J. Sun, "Shufflenet v2: Practical guidelines for efficient CNN architecture design," in *Proc. Comput. Vis.–ECCV 2018*, V. Ferrari, M. Hebert, C. Sminchisescu, and Y. Weiss, Eds., Cham, Switzerland: Springer International Publishing, 2018, pp. 122–138.

[36] Y. Jiang et al., "Identifying individual brain development using multimodality brain network," *Commun. Biol.*, vol. 7, no. 1, 2024, Art. no. 1163.

[37] T. Zhou et al., "BG-GAN: Generative ai enable representing brain structure-function connections for Alzheimer's disease," *IEEE Trans. Consum. Electron.*, vol. 71, no. 2, pp. 3087–3098, May 2025, doi: 10.1109/TCE.2025.3543943.

[38] S. G. Mueller et al., "The Alzheimer's disease neuroimaging initiative," *Neuroimaging Clin.*, vol. 15, no. 4, pp. 869–877, 2005.

[39] Z. Cui, S. Zhong, P. Xu, Y. He, and G. Gong, "Panda: A pipeline toolbox for analyzing brain diffusion images," *Front. Hum. Neurosci.*, vol. 7, no. 42, 2013, Art. no. 42.

[40] J. Wang, X. Wang, M. Xia, X. Liao, A. Evans, and Y. He, "Gretna: A graph theoretical network analysis toolbox for imaging connectomics," *Front. Hum. Neurosci.*, vol. 9, 2015, Art. no. 386, doi: 10.3389/fnhum.2015.00386.

[41] J. Yuan et al., "Graph attention transformer network for multi-label image classification," *ACM Trans. Multimedia Comput. Commun. Appl.*, vol. 19, no. 4, 2023, Art. no. 150.

[42] J. Xiao, L. Yang, and S. Wang, "Graph isomorphism network for materials property prediction along with explainability analysis," *Comput. Mater. Sci.*, vol. 233, 2024, Art. no. 112619, doi: 10.1016/j.commatsci.2023.112619.

[43] F. Salami, A. Bozorgi-Amiri, G. M. Hassan, R. Tavakkoli-Moghaddam, and A. Datta, "Designing a clinical decision support system for Alzheimer's diagnosis on Oasis-3 data set," *Biomed. Signal Process. Control*, vol. 74, 2022, Art. no. 103527, doi: 10.1016/j.bspc.2022.103527.

[44] L. V. der Maaten and G. Hinton, "Visualizing data using T-SNE," *J. Mach. Learn. Res.*, vol. 9, no. 11, pp. 2579–2605, 2008.

[45] C. J. Stam, B. F. Jones, G. Nolte, M. Breakspear, and P. Scheltens, "Small-world networks and functional connectivity in Alzheimer's disease," *Cereb. Cortex*, vol. 17, no. 1, pp. 92–99, 2007, doi: 10.1093/cercor/bhj127.

[46] A. Budson, A. Sullivan, E. Mayer, K. Daffner, P. Black, and D. Schacter, "Suppression of false recognition in Alzheimer's disease and in patients with frontal lobe lesions," *Brain*, vol. 125, no. 12, pp. 2750–2765, 2002.

[47] D. Berron, D. van Westen, R. Ossenkoppele, O. Strandberg, and O. Hansson, "Medial temporal lobe connectivity and its associations with cognition in early Alzheimer's disease," *Brain*, vol. 143, no. 4, pp. 1233–1248, 2020.

[48] A. M. Buunk et al., "Social cognition impairments after aneurysmal subarachnoid haemorrhage: Associations with deficits in interpersonal behaviour, apathy, and impaired self-awareness," *Neuropsychologia*, vol. 103, pp. 131–139, 2017, doi: 10.1016/j.neuropsychologia.2017.07.015.

[49] R. L. Buckner, "Memory and executive function in aging and ad: Multiple factors that cause decline and reserve factors that compensate," *Neuron*, vol. 44, no. 1, pp. 195–208, 2004.

[50] M. Kaiser, "A tutorial in connectome analysis: Topological and spatial features of brain networks," *NeuroImage*, vol. 57, no. 3, pp. 892–907, 2011, doi: 10.1016/j.neuroimage.2011.05.025.

[51] M. Yu, O. Sporns, and A. J. Saykin, "The human connectome in Alzheimer disease—Relationship to biomarkers and genetics," *Nature Rev. Neurol.*, vol. 17, no. 9, pp. 545–563, 2021.

Hongjie Jiang received the bachelor's degree from Sichuan University, Sichuan, China, in 2022, and the master's degree from the Southern University of Science and Technology, Shenzhen, China, in 2026. From 2023 to 2026, he conducted research training and was with the Shenzhen Institutes of Advanced Technology, Chinese Academy of Sciences, Shenzhen. His research interests include generative AI, pre-trained models, and brain network analysis.

Yongcheng Zong received the bachelor's degree from Xidian University, Shanxi, China, in 2021, and the master's degree from the University of Chinese Academy of Sciences, Beijing, China, in 2024. From 2021 to 2024, he conducted research training and was with the Shenzhen Institutes of Advanced Technology, Chinese Academy of Sciences, Shenzhen, China. His research interests include diffusion model and multimodal brain network computing.

Guoheng Huang received the Bachelor of Science degree in mathematics and applied mathematics and the Master of Engineering degree in computer science from South China Normal University, Guangzhou, China, in 2008 and 2012, respectively, and the Ph.D. degree in software engineering from the University of Macau, Macau, China, in 2017. He is currently a CCF Member and an Associate Professor of computer science with the Guangdong University of Technology, Guangzhou. His research interests include computer vision, pattern recognition, and artificial intelligence. As a key member of the Guangdong Key Laboratory of Cyber-Physical System, he has hosted and undertaken a few national and provincial-level scientific research projects, including the Natural Science Foundation of China and the National Key Research and Development Plan.

Sibo Qiao received the master's degree in software engineering and the Ph.D. degree in computer science from the China University of Petroleum, Qingdao, China, in 2020 and 2023, respectively. He is currently an Associate Professor with the School of Software, Tiangong University, Tianjin, China. His research interests include data mining, deep learning, and machine learning.

Shanshan Wang received the B.S. degree in biomedical engineering from Central South University, Hunan, China, in 2009. She is currently working toward the double-Ph.D. degree in biomedical engineering as a cotuttele student with Shanghai Jiao Tong University, Shanghai, China, and in computer science with the University of Sydney, Sydney, NSW, Australia. Her current research interests include inverse problem in medical imaging and image processing such as MR/PET image reconstruction, image denoising, and dictionary learning.

Authorized licensed use limited to: OAKLAND UNIVERSITY. Downloaded on February 19,2026 at 19:41:57 UTC from IEEE Xplore. Restrictions apply.

This article has been accepted for inclusion in a future issue of this journal. Content is final as presented, with the exception of pagination.

Peipeng Liang is currently a Professor, Doctoral Supervisor, and the Vice Dean for research with the School of Psychology, Capital Normal University, Beijing, China, and the Deputy Director of the Beijing Key Laboratory of Learning and Cognition. He was a Visiting Scholar with the Department of Psychology, Carnegie Mellon University, and The Chinese University of Hong Kong (Prince of Wales Hospital). His research interests include the neural mechanisms of reasoning, psychological assessment for proactive health, and intervention-based regulation. He is on the Editorial Board members of several international journals.

Michael Kwok-Po Ng (Senior Member, IEEE) received the B.Sc. and M.Phil. degrees from The University of Hong Kong, Hong Kong, in 1990 and 1992, respectively, and the Ph.D. degree from The Chinese University of Hong Kong, Hong Kong, in 1995. He was a Research Fellow with Computer Sciences Laboratory, Australian National University, from 1995 to 1997, and an Assistant/Associate Professor with The University of Hong Kong from 1997 to 2005. From 2006 to 2019, he was a Professor/Chair Professor with the Department of Mathematics, Hong Kong Baptist University, Hong Kong. From 2019 to 2023, he was a Chair Professor with the Research Division of Mathematical and Statistical Science, The University of Hong Kong. He is currently a Chair Professor of mathematics and data science with Hong Kong Baptist University. His research interests include bioinformatics, image processing, scientific computing, and data mining. He is selected for the 2017 Class of Fellows of the Society for Industrial and Applied Mathematics. He was the recipient of the Feng Kang Prize for his significant contributions in scientific computing. He is on the Editorial Board Members of several international journals.

Shuqiang Wang (Senior Member, IEEE) received the Ph.D. degree from the City University of Hong Kong, Hong kong. He was a Research Scientist of Huawei Technologies Noah's Ark Lab. He held a Postdoctoral fellowship with the University of Hong Kong. He is currently a Professor with the Shenzhen Institutes of Advanced Technology, Chinese Academy of Science, Beijing, China, and he is also a Research Fellow of mathematics and data science with Hong Kong Baptist University, Hong Kong. His current research interests include brain image computing, brain-computer interface, and optimization theory.

Authorized licensed use limited to: OAKLAND UNIVERSITY. Downloaded on February 19,2026 at 19:41:57 UTC from IEEE Xplore. Restrictions apply.